

Kunwar Abhuday Singh

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PROFILE

Engineer who builds things that ship — production AI agents, autonomous robots competing internationally, and open-source ML tooling with real users. Comfortable across the stack: model development and deployment, data pipelines, distributed systems, and edge inference. Motivated by hard problems with measurable output.

EDUCATION

Manipal Institute of Technology

Bachelor of Technology, Computer Science and Engineering

Manipal, India

Jul. 2023 – Mar. 2027

EXPERIENCE

KLIV Group, IIT Kharagpur

Jan. 2025 – Present

Deep Learning Research Intern

Remote

- Optimized quantized CNN and Transformer models for edge deployment on Jetson Nano and Coral EdgeTPU via INT8 quantization and knowledge distillation, achieving **35% inference speed-up** at 90%+ accuracy and **40% model size reduction**.
- Benchmarked inference throughput across ONNX Runtime, PyTorch, and Intel Neural Compressor; characterized hardware-specific failure modes (ARM ISA incompatibility, OOM behavior) used to guide deployment decisions.

Incubyte

Jun. 2025 – Jul. 2025

Software Engineering Intern

Remote

- Built and deployed **20+ enterprise AI agents** using Google Cloud and Agno framework via RESTful microservices architecture, integrated into daily production workflows.
- Automated customer support and business analytics pipelines through agent orchestration, reducing manual workload by **60%** and increasing operational throughput by **40%**.

Team RoboManipal, MIT MAHE

Mar. 2024 – Present

Robotics Software Engineer

Manipal, India

- Developed real-time 6-DOF pose estimation using ROS and OpenCV at **20+ FPS**; led ABU Robocon 2024 Nationals qualification scoring **99/100** in Stage 1. Achieved **Top 8** at RoboCup@Work Germany 2025 and **Top 10** at Brazil 2025.

PROJECTS

Advance Conv Benchmarks – Open Source Python Library | [GitHub](#) *Python, PyTorch, TensorFlow, JAX, ONNX Runtime, PyPI*

- Published [PyPI library](#) for neural network FLOP counting and inference benchmarking across PyTorch, TensorFlow, and JAX; supports custom layer architectures. Accompanying research paper in peer-review.

YOLOv8 Ensemble Object Detection & 3D Pose Extraction | [GitHub](#) *Python, PyTorch, YOLOv8, CUDA, OpenCV*

- Built distributed multi-GPU detection pipeline achieving **98%+ accuracy**; integrated depth camera fusion for millimeter-precision 3D pose estimation with **40% latency reduction** via parallel processing.

BirdCLEF Bioacoustic Species Detection | [GitHub](#) *Python, PyTorch, EfficientNet-B0, MLflow, Librosa*

- Built end-to-end audio classification pipeline on BirdCLEF 2024 dataset; engineered mel-spectrogram features from raw recordings and trained EfficientNet-B0 with MLflow experiment tracking across model variants.
- Benchmarked EfficientNet-B0 against Audio Spectrogram Transformer (AST) on imbalanced multi-class bird species data; implemented class-weighted loss and augmentation to handle real-world recording noise.

ROOT-ML Bridge – Physics & HPC Data Pipeline | [GitHub](#) *C++, Python, ROOT, RDataFrame, Parquet, XGBoost*

- Built end-to-end ML pipeline bridging CERN's ROOT framework with Python ML ecosystem; implemented memory-efficient chunked processing via RDataFrame with automated provenance tracking and full ROOT → Parquet → XGBoost round-trip.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, Java

ML & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, HuggingFace Transformers, OpenCV, NLP, Computer Vision

MLOps & Infra: ONNX Runtime, MLflow, Docker, Git, CI/CD, REST APIs, Microservices, Distributed Systems, ROS

Data & Cloud: Pandas, NumPy, SQL, GCP, Azure, MySQL, PostgreSQL, MongoDB, Parquet

Specialized: INT8 Quantization, Knowledge Distillation, Edge Deployment, Model Pruning, Pose Estimation, SLAM

RESEARCH & OPEN SOURCE

ML Benchmarking Paper: Comparing inference runtimes (PyTorch, ONNX, INC) across x86-64, Raspberry Pi 5, Jetson Nano — currently in reviewer-response phase.

CERN ROOT Contributions: Fixed silent data corruption bug in genomics tooling (ramtools PR #36); authored Issue #49 and draft PR #50 addressing hardcoded compression override.

Scientific RAG System: Co-authoring research on multi-document RAG with automated meta-analysis generation; submission in preparation.

ACHIEVEMENTS

RoboCup@Work Germany 2025: Top 8 internationally | **RoboCup@Work Brazil 2025:** Top 10 internationally | **Aether**

BytesBattle 2025: National Winners | **IISC RoboNautica 2025:** Runners-up